

resultsProposed_alpha.xlsx

Sheet: alpha

α -> Confidence level.

Column 1: alpha index.

Column 2: value.

MARKET POSITION

Sheet: mpDA

mp_t^{DA} -> Market position of the facility at the DA stage (MWh).

Column 1: time period t index.

Column 2: alpha index.

Column 3: value.

Sheet: rIB

$r_{t,s}^{IB}$ -> Realized production of the facility (MWh).

Column 1: time period t index.

Column 2: scenario s index.

Column 3: alpha index.

Column 4: value.

Sheet: dIB

$d_{t,s}^{IB}$ -> Deviation of the facility (MWh).

Column 1: time period t index.

Column 2: scenario s index.

Column 3: alpha index.

Column 4: value.

RENEWABLE FACILITY

Sheet: forecastPV

$RES_t^{R,DA}$ -> Forecast available production of RES (MW).

Column 1: time period t index.

Column 2: value.

Sheet: resRDA

$res_t^{R,DA}$ -> RES production injected into the grid at the DA stage (MW).

Column 1: time period t index.

Column 2: alpha index.

Column 3: value.

Sheet: resRIB

$res_{t,s}^{R,IB}$ -> RES production injected into the grid (MW).

Column 1: time period t index.

Column 2: scenario s index.

Column 3: alpha index.

Column 4: value.

CHARGING OF THE BATTERY

Sheet: chBDA

$ch_t^{B,DA}$ -> Battery charging power at the DA stage (MW).

Column 1: time period t index.

Column 2: alpha index.

Column 3: value.

Sheet: chBIBminus

$ch_{t,s}^{B,IB,-}$ -> Balancing power provided by battery charging; negative direction (MW).

Column 1: time period t index.

Column 2: scenario s index.

Column 3: alpha index.

Column 4: value.

Sheet: chBIBplus

$ch_{t,s}^{B,IB,+}$ -> Balancing power provided by battery charging; positive direction (MW).

Column 1: time period t index.

Column 2: scenario s index.

Column 3: alpha index.

Column 4: value.

DISCHARGING OF THE BATTERY

Sheet: disBDA

$dis_t^{B,DA}$ -> Battery discharging power at the DA stage (MW).

Column 1: time period t index.

Column 2: alpha index.

Column 3: value.

Sheet: disBIBminus

$dis_{t,s}^{B,IB,-}$ -> Balancing power provided by battery discharging; negative direction (MW).

Column 1: time period t index.

Column 2: scenario s index.

Column 3: alpha index.

Column 4: value.

Sheet: disBIBplus

$dis_{t,s}^{B,IB,+}$ -> Balancing power provided by battery discharging; positive direction (MW).

Column 1: time period t index.

Column 2: scenario s index.

Column 3: alpha index.

Column 4: value.

STATE-OF-ENERGY OF THE BATTERY

Sheet: soeBDA

$soe_t^{B,DA}$ -> Battery state-of-energy at the DA stage (MWh).

Column 1: time period t index.

Column 2: alpha index.

Column 3: value.

Sheet: soeBIB

$soe_{t,s}^{B,IB}$ -> Battery state-of-energy (MWh).

Column 1: time period t index.

Column 2: scenario s index.

Column 3: alpha index.

Column 4: value.

CONSUMPTION OF THE ELECTROLYZER

Sheet: elEDA

$el_t^{E,DA}$ -> Actual power required to produce a kilogram of hydrogen at the DA stage (MW).

Column 1: time period t index.

Column 2: alpha index.

Column 3: value.

Sheet: elEIB

$el_{t,s}^{E,IB}$ -> Actual power required to produce a kilogram of hydrogen (MW).

Column 1: time period t index.

Column 2: scenario s index.

Column 3: alpha index.

Column 4: value.

Sheet: elEIBminus

$el_{t,s}^{E,IB,-}$ -> Balancing power provided by electrolyzer; negative direction (MW).

Column 1: time period t index.

Column 2: scenario s index.

Column 3: alpha index.

Column 4: value.

Sheet: elEIBplus

$el_{t,s}^{E,IB,+}$ -> Balancing power provided by electrolyzer; positive direction (MW).

Column 1: time period t index.

Column 2: scenario s index.

Column 3: alpha index.

Column 4: value.